EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examination 2018
General Certificate of Education Advanced Level
Higher 1

CHEMISTRY

8873/01

Paper 1 Multiple Choice

24 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, civics group and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this question paper.
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **9** printed pages and **1** blank page.

1 A giant molecule contains a large amount of carbon of isotopes ^{12}C and ^{13}C. It was found that the relative atomic mass of carbon in the molecule is 12.20.

What is the ratio of ^{12}C to ^{13}C in the molecule?

A 4:1 **B** 3:1 **C** 1:3 **D** 1:4

2 A 30 cm^3 sample of butane, C_4H_{10}, was completely reacted in a limited supply of oxygen to produce 60 cm^3 of carbon dioxide and 60 cm^3 of carbon monoxide.

All volumes were measured at room temperature and pressure.

Which volume of oxygen was used?

A 90 cm^3 **B** 120 cm^3 **C** 150 cm^3 **D** 165 cm^3

3 Molybdenum can form many complex oxy-ions. It has been reported that a complex molybdenum oxyanion can self-assemble to a large doughnut-shaped structure with a 3.6 nm diameter.

The oxyanion unit has the formula $[Mo_{36}O_{112}(H_2O)_{16}]^{8-}$.

Calculate the oxidation state of molybdenum in this oxyanion unit.

A +3 **B** +4 **C** +5 **D** +6

4 Carbon monoxide reduces iodine(V) oxide to iodine.

$$5CO + I_2O_5 \rightarrow 5CO_2 + I_2$$

This reaction can be used to estimate carbon monoxide quantitatively as the liberated iodine can be reacted with sodium thiosulfate.

How many moles of thiosulfate ions would reduce the iodine produced from 1 mole of carbon monoxide in the above equation?

A 0.1 **B** 0.4 **C** 2.0 **D** 2.5

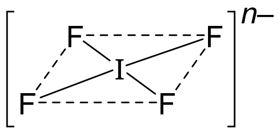
5 Which of the following particles would have a half-filled set of p orbitals on losing an electron?

A Si^- **B** P **C** P^- **D** S^+

6 Which of the following organic molecules does **not** have a permanent dipole?

A C_2Cl_4 **B** CF_2Cl_2 **C** $CHCl_3$ **D** CF_2CCl_2

7 An ion IF_4^{n-} has a square planar structure as shown below.



What is the value of *n*?

A 1 **B** 2 **C** 3 **D** 4

8 When 13.08 g of zinc was added to 250 cm^3 of 1.0 mol dm^{-3} aqueous copper(II) sulfate, the temperature of the solution rose by 15 ºC.

$$Cu^{2+}(aq) + Zn(s) \rightarrow Zn^{2+}(aq) + Cu(s)$$

What is the enthalpy change (in kJ mol^{-1}) for the above reaction?
[Specific capacity of the final solution is 4.20 J g^{-1} K^{-1}.]

A –1512 **B** –79.1 **C** –78.8 **D** –4.12

9 Which of the following will have a positive ΔH value?

1 $\frac{1}{2}O_2(g) \longrightarrow O(g)$

2 $O_2(g) + 2e^- \longrightarrow O_2^{2-}(g)$

3 $O_2(g) + O(g) \longrightarrow O_3(g)$

A **1**, **2** and **3**

B **1** and **2** only

C **2** and **3** only

D **1** only

10 The lattice energy of rubidium fluoride, RbF, is –760 kJ mol^{-1} and the lattice energy of caesium chloride, CsC*l*, is –650 kJ mol^{-1}

Which value is likely to be the lattice energy of caesium fluoride, CsF, in kJ mol^{-1}?

A –460 **B** –550 **C** –680 **D** –800

11 Consider the following equilibrium reaction:

$$3Fe(s) + 4H_2O(g) \quad Fe_3O_4(s) + 4H_2(g)$$

Which of the following gives the units for the equilibrium constant, K_c?

A mol dm^{-3} **B** mol^{-2} dm^{6} **C** mol^{4} dm^{-12} **D** no units

12 Two students set up the equilibrium system below.

$$CH_3COOC_2H_5(l) + H_2O(l) \quad C_2H_5OH(l) + CH_3COOH(l)$$

The students titrated samples of the equilibrium mixture with NaOH(aq), to determine the concentration of CH_3COOH.

The students used their results to calculate a value for K_c, but their values were different.

Which of the reason(s) below could explain why the calculated values for K_c were different?

1 Each student carried out their experiment at a different temperature.

2 Each student used a different concentration of NaOH(aq) in their titration.

3 Each student titrated a different volume of the equilibrium mixture.

A **1**, **2** and **3**

B **1** and **2** only

C **2** and **3** only

D **1** only

13 20.0 cm^3 of 0.10 mol dm^{-3} sulphuric acid, H_2SO_4, was mixed with 20.0 cm^3 of 0.10 mol dm^{-3} aqueous sodium hydroxide. The following reaction occurs

$$H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4\,(aq) + 2H_2O(l)$$

What is the pH of the resulting solution?

A 1.0 **B** 1.3 **C** 1.6 **D** 7.0

14 The hydrolysis of sucrose in aqueous solution is catalysed by hydrogen ions, such as from hydrochloric acid.

Which of the following procedures can be used to determine the order of the reaction with respect to hydrogen ions?

A Measure the rate of the reaction several times, but with a different concentration of hydrochloric acid each time.

B Add a suitable indicator and watch for the time when the colour changes.

C Remove samples at various time intervals and titrate against a standard solution of sodium hydroxide.

D Measure the change in pH during the reaction.

15 Consider the oxides of the Period 3 elements.
Which property decreases from Na_2O to SiO_2 and also from SiO_2 to P_4O_{10}?

A covalent character

B melting point

C pH when mixed with water

D solubility in aqueous alkali

16 Aluminium is the third most abundant element in the Earth's crust.

Which of the following statements is **not** true of the element and its compounds?

A Aluminium chloride exists as a dimer in solid state.

B A solution of aluminium chloride turns blue litmus red.

C Aluminium oxide dissolves in both aqueous acids and aqueous alkalis.

D Aluminium fluoride has a much higher melting point than aluminium bromide.

17 Element **G** is found in Period 3 of the Periodic Table.

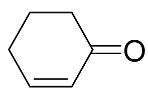
The oxide and chloride of element **G** are separately mixed with water. The two resulting solutions have the same effect on litmus paper.

What is the identity of element **G**?

A aluminium **B** magnesium

C phosphorus **D** sodium

18 How many σ and π bonds are there in a molecule of the cyclohex-2-enone?



cyclohex-2-enone

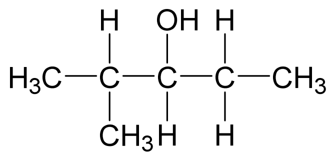
	σ	π
A	5	4
B	7	2
C	8	2
D	15	2

19 The alkynes are a series of hydrocarbons containing one C≡C triple bond per molecule. They have the general formula C_nH_{2n-2}.

How many **structural** isomers are there for the alkyne containing six carbon atoms per molecule?

A 3 **B** 5 **C** 6 **D** 7

20 How many different alkenes, including geometrical isomers, could be produced when



reacts with hot concentrated sulfuric acid?

A 2 **B** 3 **C** 4 **D** 5

21 Compound **X** has the molecular formula $C_4H_{10}O_2$.

X has an unbranched carbon chain and contains two OH groups.

On reaction with an excess of hot, acidified, aqueous manganate(VII) ions, **X** is converted into a compound of molecular formula $C_4H_6O_4$.

To which two carbon atoms in the chain of **X** are the two OH groups attached?

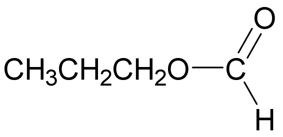
A 1st and 2nd

B 1st and 3rd

C 1st and 4th

D 2nd and 3rd

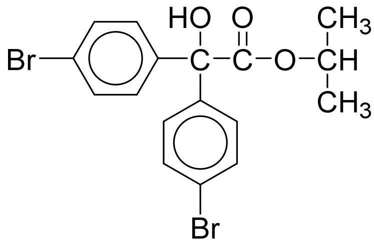
22 The structural formula of a compound **X** is shown below.



What is the name of compound **X** and how does its boiling point compare with that of butanoic acid?

	name of **X**	boiling point of **X**
A	methyl propanoate	higher
B	methyl propanoate	lower
C	propyl methanoate	higher
D	propyl methanoate	lower

23 *Acarol* is sold as an insecticide for use on fruit and vegeta



Acarol

The final stage in its manufacture is an esterification reaction.

Which alcohol is used in the final stage to form *Acarol*?

A propan-2-ol

B propan-1-ol

C methanol

D di(4-bromophenyl)methanol

24 Propan-1-ol is heated with Al_2O_3. The organic product is then reacted with bromine.
What is the final product/s of these two reactions?

A 1-bromopropane

B 1-bromopropane and 2-bromopropane

C 1,2-dibromopropane

D 1,3-dibromopropane

25 A bromine-containing organic compound, **T**, undergoes an elimination reaction when treated with hot ethanolic sodium hydroxide solution.

Which compound could be **T**?

A CH_3Br

B C_2Br_6

C $CH_3CH_2CBr_3$

D $(CH_3)_2C{=}CBr_2$

26 "Clearfilm" is manufactured from a polymer made by copolymerising $CH_2{=}CHCl$ with $CH_2{=}CCl_2$ in a regular 'head to tail' linkage, where CH_2 is taken as the 'head' of each monomer.

Which of the following could represent parts of the polymer chain in "Clearfilm'

1 —$CHCl$—CH_2—CCl_2—

2 —CH_2—$CHCl$—CH_2—

3 —CH_2—$CHCl$—CCl_2—

A **1** and **2** only

B **1** and **3** only

C **2** and **3** only

D **1**, **2** and **3**

27 Nylon 6 has the following formula and undergoes acidic hydrolysis.

$$-\!\!\left(NH(CH)_5CO\right)_n\!\!-$$

What is the product of the acidic hydrolysis of Nylon 6?

A $HO_2C(CH_2)_4CO_2H$

B $H_3\overset{+}{N}(CH_2)_6\overset{+}{N}H_3$

C $H_3\overset{+}{N}(CH_2)_5CHO$

D $H_3\overset{+}{N}(CH_2)_5CO_2H$

28 Poly(ethene) and PVC are examples of addition polymers.
Which statements are correct?

1 On combustion, PVC can produce carbon monoxide, carbon dioxide and hydrogen chloride.

2 When poly(ethene) is buried in a landfill site, it will biodegrade when in contact with bacteria.

3 The empirical formula of an addition polymer is the same as that of the monomer.

A **1** and **2** only

B **1** and **3** only

C **2** and **3** only

D **1**, **2** and **3**

29 Chemists have recently synthesised the smallest "beakers" for carrying out chemical reactions. The "beakers" are the junctions from a network of hollow polymer nanofibres. The volume of the beakers is about 4×10^{-18} dm^3.

A "beaker" is full of a solution of glucose of concentration 5×10^{-4} mol dm^{-3}.

Calculate the number of glucose molecules in the "beaker".

A 602 **B** 1204 **C** 1806 **D** 2408

30 Which of the following uses are attributed to the high surface to volume ratio of nanoparticles?

1 The use of graphene in tennis rackets.

2 The use of silver nanoparticles to treat wounds.

3 The use of platinum nanoparticles as catalyst in catalytic converter in cars.

4 Colloidal solution of silver nanoparticles is used in painting as it appears yellow.

A **1**, **2** and **3** only

B **2**, **3** and **4** only

C **2** and **3** only

D **3** and **4** only

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Answer :

Question	Answer	Question	Answer
1	**A**	16	**A**
2	**D**	17	**C**
3	**D**	18	**D**
4	**B**	19	**D**
5	**C**	20	**B**
6	**A**	21	**C**
7	**A**	22	**D**
8	**C**	23	**A**
9	**B**	24	**C**
10	**C**	25	**C**
11	**D**	26	**A**
12	**D**	27	**D**
13	**B**	28	**B**
14	**A**	29	**B**
15	**C**	30	**B**